

Name: _____

Solutions

Date: _____

Year 10 Mathematics

Test 1, 2015

Topics – Surface Area and Volume



SAFETY BAY
SENIOR HIGH SCHOOL

imagine believe achieve

_____ 50 _____ %
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Total Time: 55 minutes

Weighting: 5% of the year.

Equipment Allowed:

Page notes (A4 one side), Calculator

1. [4 marks: 1, 1, 1, 1]

Convert the measurements below

a. 0.0059 m² to cm²

59 cm² ✓

b. 4.9 kL to L

4900 L ✓

c. 720 ha to square kilometres

7.2 km² ✓

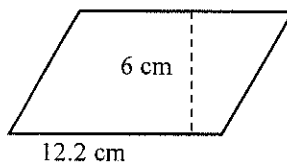
d. 320 m³ to ML

0.32 ML ✓

2. [4 marks: 2, 2]

Calculate the area of the following shapes (each 1 dp)

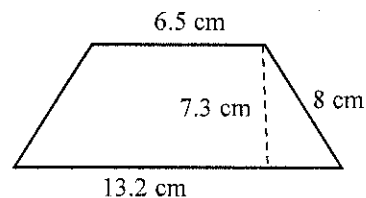
a.



$$\begin{aligned} \text{Area} &= 6 \times 12.2 \\ &= 73.2 \text{ cm}^2 \end{aligned}$$

✓✓

b.



$$\begin{aligned} \text{Area} &= \frac{(6.5 + 13.2)}{2} \times 7.3 \\ &= 71.9 \text{ cm}^2 \end{aligned}$$

✓✓

3. [2 marks: $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{1}{2}$]

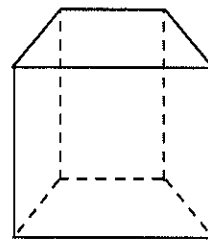
Identify the following 3D shapes

a.



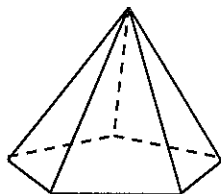
Cylinder $\frac{1}{2}$

b.



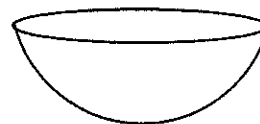
Trapezoid Prism $\frac{1}{2}$

c.



Pentagonal Pyramid $\frac{1}{2}$

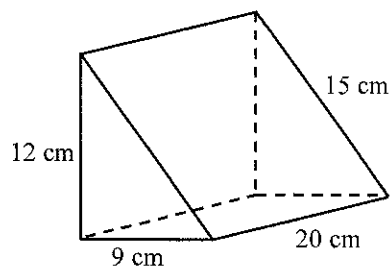
d.



Hemisphere $\frac{1}{2}$

4. [7 marks: 5, 2]

Use the following shape to answer the questions



a.

Calculate the surface area

b.

Calculate the volume

$$\Delta = \frac{1}{2} \times 12 \times 9 = 54 \times 2 = 108 \checkmark$$

$$\text{Back } \square = 12 \times 20 = 240 \checkmark$$

$$\text{Bottom } \square = 9 \times 20 = 180 \checkmark$$

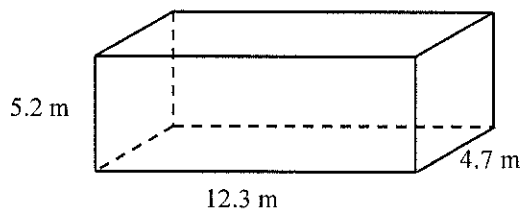
$$\text{Front } \square = 15 \times 20 = \underline{300} \checkmark$$

$$\text{Total Surface Area} = 828 \text{ cm}^2 \checkmark$$

$$\begin{aligned} \text{Volume} &= \frac{1}{2} \times 12 \times 9 \times 20 \checkmark \\ &= 1080 \text{ cm}^3 \checkmark \end{aligned}$$

5. [3 marks]

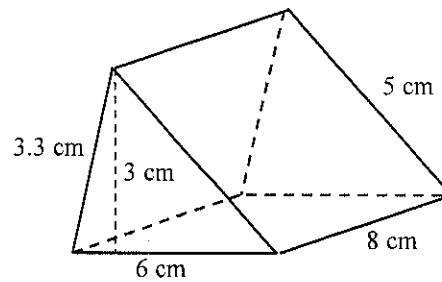
Calculate the surface area of the following cuboid



$$\begin{aligned}
 \text{Front/Back} &= 12.3 \times 5.2 \times 2 = 127.92 \\
 \text{Left/Right} &= 5.2 \times 4.7 \times 2 = 48.88 \quad \checkmark \\
 \text{Top/Bottom} &= 12.3 \times 4.7 \times 2 = 115.62 \quad \checkmark \\
 \text{Total Surface Area} &= 292.42 \text{ m}^2 \quad \checkmark
 \end{aligned}$$

6. [7 marks: 5, 2]

Use the following shape to answer the questions



a. Calculate the surface area

b. Calculate the volume

$$\begin{aligned}
 2\Delta &= 3 \times 6 \times \frac{1}{2} \times 2 = 18 \quad \checkmark \\
 \text{Back} &= 3.3 \times 8 = 26.4 \quad \checkmark \\
 \text{Bottom} &= 6 \times 8 = 48 \quad \checkmark \\
 \text{Front} &= 5 \times 8 = 40 \quad \checkmark \\
 \text{Total Surface Area} &= 132.4 \text{ cm}^2 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{Volume} &= 3 \times 6 \times \frac{1}{2} \times 8 \\
 &= 72 \text{ cm}^3 \quad \checkmark \checkmark
 \end{aligned}$$

7. [4 marks]

What area of tin is needed to make an open box which is 4 cm deep with a base measuring 8 cm by 6 cm?

$$\begin{array}{rcl}
 \text{Bottom} & = & 8 \times 6 = 48 \checkmark \\
 \text{Front/Back} & = & 8 \times 4 \times 2 = 64 \checkmark \\
 \text{left/Right} & = & 6 \times 4 \times 2 = 48 \checkmark \\
 & & \hline
 & & 160 \text{ cm}^2 \checkmark
 \end{array}$$

8. [9 marks: 6, 3]

- a Use Heron's formula to help you find the surface area of a triangular prism if the triangular ends have sides lengths measuring 6m, 7m and 8m and the prism is 11m long. If necessary, round your answer to two decimal places.

$$\begin{aligned}
 \Delta &= 2 \times \sqrt{s(s-a)(s-b)(s-c)} \\
 &= 2 \times \sqrt{10.5 \times 4.5 \times 3.5 \times 2.5} \checkmark \\
 &= 2 \times 20.333 \\
 &= 40.67 \checkmark
 \end{aligned}$$

$$\begin{aligned}
 s &= \frac{1}{2}(a+b+c) \\
 &= \frac{1}{2}(6+7+8) \\
 &= 10.5 \checkmark
 \end{aligned}$$

Rectangle

$$\begin{array}{rcl}
 1. & 6 \times 11 & = 66 \\
 2. & 7 \times 11 & = 77 \checkmark \checkmark \\
 3. & 8 \times 11 & = 88
 \end{array}$$

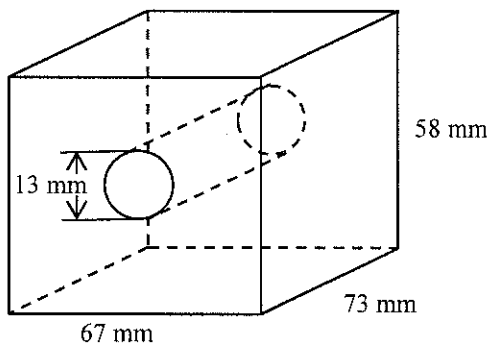
$$\text{Total Surface Area} = 271.67 \text{ m}^2 \checkmark$$

- b Use your calculations from above to calculate the volume of the prism (to 1 dp)

$$\begin{aligned}
 \text{Volume} &= 20.333 \times 11 \checkmark \\
 &= 223.7 \text{ m}^3 \checkmark \checkmark
 \end{aligned}$$

9. [5 marks]

Calculate the volume of the following shape to the nearest mm^3



Volume of cuboid

$$= 67 \times 73 \times 58$$

$$= 283678 \text{ mm}^3 \quad \checkmark$$

Volume of cylinder

$$= \pi \times 6.5^2 \times 73$$

$$= 9689.457 \quad \checkmark$$

$$\text{Total Volume} = 283678 - 9689.457 \quad \checkmark$$

$$= 273989 \text{ mm}^3 \quad \checkmark \checkmark$$

10. [5 marks]

The volume of a square based pyramid is given as 600 m^3 and it has a perpendicular height of 12.5 m . Calculate the surface area of one of its sides to the nearest m^2 .

$$\text{Volume} = \frac{1}{3} \times \text{base} \times \text{height}$$

$$600 = \frac{1}{3} \times \text{base} \times 12.5$$

$$\text{base} = \frac{600 \times 3}{12.5}$$

$$= 144 \text{ m}^2 \quad \checkmark \checkmark$$

$$\text{length of one side} = 12 \text{ m}$$

Diagonal side of Pyramid

$$= \sqrt{6^2 + 12.5^2} \quad \checkmark$$

$$= 13.865 \text{ (3 dp)}$$

Area of one side

$$= \frac{1}{2} \times 12 \times 13.865$$

$$= 83 \text{ m}^2 \quad \checkmark \checkmark$$

End of Test